MILITARY SPECIFICATION

DISPENSER; DEPTH CHARGE, AERO 1A

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the Depth Charge Dispenser Aero 1A, for use on Military aircraft, which is capable of carrying and launching depth charge (practice) Mark 15 Mod 1, and depth charges Mark 15 Mod 8, and Mark 15 Mod 12.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-E-7894	Electric Power, Aircraft Characteristic of
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection For Aircraft Electronic Equipment, Format for
MIL-D-18047	Design for Name Plates for Armament Equipment

Bureau of Aeronautics

SAR-312	Aircraft Armament Changes and Bulletins; Preparation of

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STANDARDS

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Air Force - Navy Aeronautical

AND 10461	Connections - Input Power Airborne Electronic Equipment; 28 Volt DC, 115 Volt Single Phase AC, and/or 115/200 Volt Three Phase AC

Bureau of Aeronautics

56A189ASK1	List of drawings (and all drawings and specifications listed thereon)

PUBLICATIONS

Air Force - Navy Aeronautical

Bulletin No. 143	Specifications and Standards; Use of
Bulletin No. 391	Changes; Engineering, to Aircraft Engines, Propellers, and Aeronautical Equipment in Production and Service.

Bureau of Aeronautics

NAVAER 00-25-544	List of Military Specifications and Standards used by the Bureau of Aeronautics

2.2 Availability of Documents:

(1) Refer to both title and number when requesting specifications, standards, drawings and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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3. REQUIREMENTS

3.1 Materials - Materials used in the manufacture of the equipment shall conform to the applicable specifications specified herein or as listed on the applicable drawings, of Drawing 56A189ASK1, if such materials are available. Specifications and standards for all materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143. Specifications and standards shall be selected from those listed in NAVAER 00-25-544 except as otherwise provided herein. Materials conforming to contractor's specifications may be used provided the specifications are released by the Bureau of Aeronautics, are equivalent to the Government specifications, and contain requirements for adequate tests of the material. The use of contractor's specifications shall not constitute a waiver of Government inspection. All materials shall be of the lightest practicable weight and entirely suitable for the intended purpose. When materials are used which are procured under commercially-recognized non-Government specifications, a certificate, acceptable to the Government Inspector, showing compliance with such specifications, shall be furnished by the supplier of such material.

3.2 Conformance to Drawings - No departure from or revision or change to Drawing 56A189ASK1 shall be permitted without prior written approval by the Bureau of Aeronautics. In the event changes are approved, the contractor shall furnish the Bureau of Aeronautics with reproducible drawings covering the change. To obtain Bureau of Aeronautics' approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for changes either recommended to the Bureau of Aeronautics or proposed by the contractor. When required, the contractor also shall prepare an Aircraft Armament Change (AAC) and an Aircraft Armament Bulletin (AAB) to cover the approved change. ECP's shall be prepared and submitted in accordance with ANA Bulletin No. 391. AAC's and AAB's shall be prepared and submitted in accordance with Specification SAR-312 (see 6.1).

3.3 Manufacturing and Finish - Manufacturing processes and finishes shall be as shown on the applicable drawings listed on Drawing 56A189ASK1.

3.4 Standard Parts - Where applicable, utility parts such as bolts, nuts, rivets, and washers shall conform to MIL Standards or shall be interchangeable functionally and dimensionally with or replaceable by MIL standard parts without modification and shall be designated by their MIL Standard part numbers. Drawings furnished by the contractor shall call out equivalent MIL Standard part numbers and note substitutions.

3.5 Interchangeability - All parts having the same part number shall be completely interchangeable with each other with respect to installation and performance. Parts joined together as by welding or riveting shall be interchangeable as assemblies. Proper gaging methods and facilities shall be maintained to insure that this requirement is fulfilled. The interchangeability requirements between like assemblies, subassemblies, and replaceable parts as specified herein shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose.

3.6 Performance - The equipment shall be capable of carrying and launching nine Mark 15 Mod 1, nine Mark 15 Mod 8, or nine Mark 15 Mod 12 depth charges from each side of the dispenser (total 18 stores). The equipment shall be capable of launching single stores from either side or from both sides simultaneously as selected in flight. The equipment shall be capable of launching stores satisfactorily during flight under the positive n_z design limit-load conditions. The minimum launching rate for each side of the fully loaded equipment shall be three stores per minute. In case of emergency, the equipment shall be capable of jettisoning all remaining stores in one operation.

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3.6.1 Operating Life -

3.6.1.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 5,000 releases without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.6.1.2 Total Operating Life - The equipment shall have a minimum total operating life of 25,000 releases with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.6.2 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload caused by any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permament damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.7 Design and Construction - The equipment shall be in accordance with the approved drawings and the detail, operational, and performance requirements specified herein. (See 3.2)

3.7.1 Ejector Shaft Gear - When the ejector feet are in the uncocked position, the gears on the ejector shafts shall be located to mesh smoothly with the cam gears when the cam gears are rotated by the motor.

3.7.2 Motor Switch Cam - The cams on the motor shafts shall be adjusted to actuate the motor microswitches and stop the motors when the gears are unmeshed and the sear stops are held in the cocked position by the sears.

3.7.3 Jettison Motor Stop Cam - The cams on the ejector shafts shall be adjusted to allow the jettison motor stop micro switches to remain in the closed position when the ejector shafts are in the cocked position. When the ejector shafts are released, the cams shall actuate the switches to the open position.

3.7.4 Jettison Cables - The jettison cable shall be adjusted so that actuation of the solenoid rotates the jettison stop clear of the sear stop.

3.7.5 Nameplate - A metal nameplate conforming to Specification MIL-D-18047 shall be attached securely to the forward face of the equipment.

3.7.6 Weight - The overall weight of the equipment shall not exceed 62 pounds.

3.7.7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be accomplished easily.

3.7.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which are to be attached to the cables shall not be supplied as part of the equipment.

3.7.9 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

DC Power 28 Volts System

Operating Voltage Limits	20 to 29 Volts
Current	20.5 Amps

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3.7.10 Input Power Connection - Input power connections shall be in accordance with Drawing AND 10461.

3.7.11 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.7.12 Strength - The loading conditions specified herein are defined by parameters such as load factors, accelerations, forces, configurations, and weights. For conditions in which forces are not completely specified to the extent necessary for the equipment to be in complete translational and rotational equilibrium, additional forces which are determined by a rational method or which are conservative shall be assumed to act in a manner such that the equipment is in equilibrium.

3.7.12.1 Limit Loads - The loads resulting from the loading conditions herein are limit loads. The limit loads shall be those developed when the equipment is loaded with stores and subjected to the accelerations specified in 3.8.10 plus the loads resulting from operation of the equipment.

3.7.12.2 Factors of Safety - The factors of safety are those specified for design loads.

3.7.12.2.1 Yield - The minimum yield factor of safety shall be 1.15 and shall be used to determine the design-yield loads.

3.7.12.2.2 Ultimate - The minimum ultimate factor of safety shall be 1.5 and shall be used to determine the design-ultimate loads.

3.7.12.3 Ultimate Strength - Failure shall not occur under design-ultimate loads resulting from the limit loads and ultimate factor of safety specified herein.

3.7.12.4 Deformations Resulting from Design-Yield Loads - Deformations occuring when the design-yield loads are applied to the equipment shall not adversely affect the mechanical operation of any part of the equipment.

3.8 Service Conditions - The equipment shall function properly after being subjected to any or all of the environmental conditions specified herein.

3.8.1 Temperature - The temperature range shall be from -62° to +71°C and may remain constant or vary at a rate as high as 1°C per second.

3.8.2 Humidity - The humidity range shall be from 0 to 100% including conditions wherein condensation takes place in the form of both water and frost.

3.8.3 Altitude - The altitude range shall be from sea level to 70,000 feet, the pressures ranging from 30 inches of mercury to 1.32 inches of mercury, and the pressure may remain constant or vary at a rate as high as 0.5 inches of mercury per second. The equipment shall operate satisfactorily at altitudes ranging from sea level to 6,000 feet, the pressure ranging from 30 to 23.98 inches of mercury remaining constant or varying as specified herein.

3.8.4 Salt Spray - The equipment shall operate satisfactorily during exposure to salt sea atmosphere.

3.8.5 Fungus - The equipment shall be fungus-proofed to resist the fungus growth conditions encountered in tropical climates.

3.8.6 Sand and Dust - The equipment shall operate satisfactorily under the sand and dust conditions encountered in desert areas.

3.8.7 Explosive Atmosphere - The equipment shall operate satisfactorily in an explosive vapor within or surrounding the equipment.

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3.8.8 Vibration - The vibration frequency shall vary from 5 to 500 cycles per second, at a double amplitude of 0.080 inch or an applied acceleration of ± 10 g, whichever is the limiting value. The equipment shall operate satisfactorily under vibration conditions. (See 4.5.3)

3.8.9 Shock - The equipment shall remain in place and shall retain all the stores loaded therein when subjected to 30 g shocks for a duration of 11 ± 1 milliseconds. The equipment is not required to operate during or after being subjected to 30 g shock. Bending and distortion will be permitted.

3.8.10 Accelerations - The accelerations shall be those experienced during flight, catapult take-off, and arrested landing of Naval aircraft as shown in Table 1.

Table 1. Design Limit Load Factors

	N_z	N_x	N_y	$\ddot{\theta}$	$\ddot{\psi}$
Flight	-3	±2	±1.5	±6	0
	+7.5				
Arrested Landing	-1.5	+8	±1.5	±12	±6
	-3.5	+2			
	-3.5	-2			
	+7	-2			
	+7	+2			
	+3	+8			
Catapult Take-off	-1	+2	±1.5	±12	±4
	-1	-7.5			
	+2	-7.5			
	+2	+2			

3.8.10.1 Load Factor, n - The load factor, n, is the ratio of a given load to an arbitrary reference load. The arbitrary reference load is the weight of the equipment loaded with stores. The subscript indicates the direction in which the equipment is accelerating.

3.8.10.2 Limit Load Factor - The limit load factor establishes a strength level for design of the equipment.

3.8.10.3 Coordinate System - Load factors with subscript, x, are positive in the aft direction. Load factors with subscript y, are positive to the left. Load factors with subscript, z, are positive upward. The three axes are mutually perpendicular.

3.8.10.4 Load Factor, $\ddot{\theta}$ - Load factor, $\ddot{\theta}$, is the pitching acceleration in radians per second per second.

3.8.10.5 Load Factor, $\ddot{\psi}$ - Load factor, $\ddot{\psi}$, is the yawing acceleration in radians per second per second.

3.8.10.6 Operation - The equipment shall operate satisfactorily after being subjected to design-yield load but is not required to operate after being subjected to loads in excess of design-yield load.

3.9 Workmanship - All workmanship shall be of a quality adequate to insure durability and proper operation, and shall be satisfactory to the Bureau of Aeronautics in all respects. Manufacturing processes and finishes shall be as specified in 3.3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - The inspection and testing of the equipment are classified as follows:

(1) Preproduction Tests - Preproduction Tests are those tests conducted on an equipment representative of the equipments to be supplied under the contract for the purpose of determining that the equipment meets all the requirements of the applicable specifications.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on equipment submitted for acceptance under the contract.

(3) Special Tests - Special Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics.

4.2 Preproduction Tests - Preproduction Tests shall be accomplished by Contractor Demonstration Tests and Service Approval Tests.

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more equipments as necessary to determine that the model as proposed by the contractor will meet all the requirements of the applicable specifications and the contract. Contractor Demonstration Tests shall be conducted in accordance with the approved test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the equipment for Service Approval Tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having laboratory facilities to conduct all required tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

4.2.2 Service Approval Tests - The Service Approval Sample shall consist of three equipments representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics. When contractor's demonstration tests are supervised, the Bureau of Aeronautics may, when conducting Service Approval Tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting Service Approval Tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. The preproduction equipments shall be tested in accordance with 4.5. An appropriate test schedule shall be arranged whereby the three service approval equipments will be undergoing tests simultaneously. Each test shall be conducted on at least one preproduction equipment.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance under the contract. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Bureau of Aeronautics shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required by this specification and signed by

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an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipments as specified in 4. 3. 2. 1.

4. 3. 1 Individual Test - Each equipment submitted for acceptance shall be subjected to the Individual Test to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall have passed the following tests successfully:

Test	Paragraph
Examination of Product	4. 5. 2
Operational	4. 5. 4

4. 3. 2 Sampling Tests -

4. 3. 2. 1 Sampling Test Schedule - The number of equipments to be selected at random from each production lot shall be as specified below. The samples furnished for sampling tests shall be in addition to the quantity of equipments on contract.

Quantity of Equipments in Production Lot	Quantity of Samples to be Selected
99 or less	1
100 to 299	3
300 to 500	6
Over 500	(see 4. 3. 2. 2)

4. 3. 2. 2 Limitation of Lot Size - For sampling purposes, lots of over 500 units shall be divided into smaller lots in accordance with the above table.

4. 3. 2. 3 Scope of Tests - Each production lot sample shall be subjected to the individual tests, (see 4. 3. 1). One sample from each production lot also shall be subjected to the following tests:

Test	Paragraph
Vibration Test	4. 5. 3(10)
Acceleration Test (To limit load in the critical condition)	4. 5. 3(12)

4. 4 Special Test - Special Tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. Equipment selected for Special Test may be selected from equipment previously subjected to the Sampling Test.

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4.4.1 Special Test Schedule - Special Tests shall be conducted on the equipment when required by the following:

(1) After an engineering change.

(2) Whenever failure reports or other information indicate additional tests are necessary.

4.4.2 Scope of Test - Special Tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the Preproduction and Sampling Tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests.

4.5 Test Procedures: -

4.5.1 General Test Requirements -

4.5.1.1 The detail description of procedures used for conducting Preproduction Tests, and Acceptance Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.5.1.2 Auxiliary equipment used in connection with these tests shall be of standard or approved type used in service installation of the test equipment and whose performance characteristics are known.

4.5.1.3 Pre-Test Check - All movable or adjustable working parts and limit switches of the equipment shall be operated sufficiently, prior to testing, to insure that they are in proper adjustment and working condition.

4.5.2 Examination of Product -

4.5.2.1 Individual Tests - Each equipment shall be examined to determine conformance with the following:

(1) Dimensions - as shown on the applicable drawings. (This may be done prior to or after assembly, as necessary.)

(2) Surface finishes

(3) Assembly

(4) Workmanship

4.5.2.2 Preproduction and Sampling Tests - In addition to the requirements of 4.5.2.1, equipments submitted for preproduction and sampling tests shall be subjected to a complete material and process check to determine conformance with 3.1, 3.3 and 3.7.6.

4.5.3 Environmental Tests - The equipment shall be subjected to the following tests in accordance with the requirements of Specification MIL-E-5272, except as otherwise specified hereinafter:

(1) High Temperature Tests

(2) Low Temperature Tests, Procedure I

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(3) Temperature Shock Tests, Procedure II

(4) Humidity Tests, Procedure III

(5) Altitude Tests, Procedure I

(6) Salt Spray Tests

(7) Fungus Resistance Tests

(8) Sand and Dust Tests, Procedure II

(9) Explosion Proof (Aeronautical) Tests, Procedure I

(10) Vibration Tests, Procedure I

(11) Shock Tests, Procedure I - The equipment loaded to capacity with Mk 15 Mod 12 depth charges shall be subjected to three impact shocks to the following load factors:

Vertically down	30 g
Transverse	15 g
Longitudinally forward	30 g

There shall be no inadvertent release of the stores or mechanical failures due to the applied shocks. (See 3. 8. 9).

(12) Acceleration Test, Procedure I and the following:

(a) Sequence of Tests - Acceleration tests, wherein failure would occur resulting in damage and repair or replacement of the test equipment or that would influence the validity of subsequent tests, shall be conducted in such sequence that:

1. Stores release tests under flight design limit-load are completed prior to conducting tests for any other conditions to loads in excess of design yield loads.

2. Design-ultimate load tests for flight conditions are completed prior to conducting tests of other-than-flight conditions to design ultimate loads.

3. All design ultimate load tests are completed prior to conducting failing load tests for any condition.

(b) Use of Test Parts - Unless proof-load tests only are required or unless specifically approved by the Bureau of Aeronautics, parts of the strength-test structure which have been subjected to structural tests shall not be used on a flight equipment.

(c) Distribution of Loads - The distribution of loads during test shall duplicate the actual distribution as closely as is practicable, except as specified in 4. 5. 3 (12) (d).

(d) Simplification and Combination of Loading - Loading conditions may be simplified by modifying the distribution of the loads applied to regions of a structure that are not critical in the loading condition being simulated in the test, or that are similar in construction to other regions of the structure that are more highly stressed during the same or other test condition. However, simplification of loading

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shall not be such that permanent deformations or failures occur because of peculiarities in the method of loading. Where practicable, one or more loading conditions may be applied simultaneously to different portions of the structure provided that the interaction of the separate loading conditions does not affect the critical design loading on any portion of the structure.

(e) Strain measurements - Strain measurements necessary to verify assumptions on which the calculations of loads on and in the structure depend, to demonstrate compliance with applicable strength requirements, and as necessary to detect incipient premature failure, shall be made during this test. Strain measurements shall be obtained at a sufficient number of load increments below limit load to establish the rate of deflection, strain, and permanent set. If a change in rate occurs, the subsequent increments shall be small enough to determine if this change is linear or curvilinear. Strain measurements may be discontinued after design ultimate load has been reached.

(f) Load increments - Up to limit load, the test loading shall be applied in increments of not more than 20 % of limit load. Test loads between limit and 150 % of the limit loads shall include design-yield load, and the load increments shall not be greater than 10% of limit load.

(g) Determination of Yield-Strength Load - The yield-strength load for each structural part shall be determined during each test of that part, except that the yield-strength load need not be determined if the structure withstands design-ultimate load without yielding.

(h) Failing Load - Failing-load tests shall be conducted until a major failure occurs in important structural members. Reinforcements representative of those that could be incorporated in flight equipments shall be made as necessary to establish reasonable structural continuity until major failures occur during these tests. If, during the course of conducting a required failing load test for a design condition selected as critical, a load equal to 125% of the calculated ultimate strength is attained without failure, the test shall be discontinued, and in lieu thereof, a test shall be conducted to failure in the next most critical design condition. This procedure shall apply only where a second design condition exists which is critical for a portion of the structure being tested. Otherwise the test shall be conducted to failure.

(i) Load Application - The equipment shall be mounted on the test apparatus so that the resultant of the acceleration of gravity and the acceleration of the centrifuge will act on the equipment in the in the same direction and with the same magnitude required in 3. 7. 12 and 3. 8. 10.

4. 5. 4 Operational Tests - The full capacity of Mark 15 Mod 12 stores (inert) shall be loaded in the equipment and successfully launched electrically a minimum of three times. A full capacity of stores shall be jettisoned successfully both electrically and manually.

4. 5. 5 Life Tests - One side of the equipment shall be loaded to full capacity of Mark 15 Mod 12 depth charges (inert) and launched electrically until the equipment has been subjected to 5,000 releases. Minor repairs and adjustments may be made during these tests to complete the required number of releases. However, there shall be no failure of a major part (base casting, ejector feet, etc.) (see 6. 2).

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4. 5. 5. 1 Test Data - The contractor shall keep a daily record as to the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures during this test, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The part function

(3) Name of the manufacturer

(4) Nature of the failure

(5) The number of releases during which the part operated prior to failure.

In the event of a part failure the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4. 5. 5. 2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be allowed to return to room temperature before continuing the test.

4. 5. 6 *Interference Tests - Interference tests shall be conducted in accordance with* Specification MIL-I-6181.

4. 6 Rejection and Retest - Any equipment failing to pass the tests specified herein shall be rejected. Equipment which has been rejected may be reworked or have parts replaced to correct defects and resubmitted for testing. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects shall be furnished to the Bureau of Aeronautics.

4. 6. 1 Sampling Inspection - If any sample equipment fails to pass the tests required in 4. 3. 2. 3, the entire lot represented by that sample shall be rejected. Rejected lots may be reworked and resubmitted for retest as specified hereinafter, subject to the approval of the Government Inspector.

4. 6. 1. 1 First Retest - The entire lot shall be reworked and resubmitted by the contractor. Twice as many sample equipments as originally submitted for the lot shall be selected at random by the Government Inspector and subjected to the tests specified in 4. 3. 2. 3. If any sample fails, the entire lot shall be rejected.

4. 6. 1. 2 Second Retest - If the failure of the first retest samples is caused by a remediable defect, the contractor may correct the defect in each equipment and again submit the lot for a second retest. The second retest shall be as specified in 4. 6. 1. 1, except that the Government Inspector shall select three times as many samples as originally submitted for the lot. If these samples fail, the lot shall be rejected. Any further rework and retest shall be approved by the Bureau of Aeronautics.

4. 7 Salvaged Pieces - Unless otherwise specified herein, any piece or pieces from an equipment may be salvaged for use, providing each salvaged piece is inspected for and is in complete compliance with the drawings and requirements of this specification.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555 and MIL-STD-129. (See 6.4).

6. NOTES

6.1 Adherence to Drawings and Specifications - Conflict Elimination - Insofar as the contractor is required by the provisions of this specification to adhere rigidly to Drawing 56A189ASK1, which also may make it impossible in some instances to comply with the performance and test requirements specified herein, then, in that event, the contractor should notify the Bureau of Aeronautics promptly of such facts and appropriate changes will be made in the applicable drawings, parts, or performance requirements by the Bureau of Aeronautics.

6.2 Availability of Dummy Stores for Test - The contractor shall submit requests to the Bureau of Aeronautics for dummy and/or inert stores to be used for testing as required by this specification.

6.3 Information Required with Bids - Each bid shall be accompanied by a statement as to whether or not any exceptions are being taken to the requirements shown in the drawings and specifications. If exceptions are being taken or deviations are anticipated, alternative proposals shall be indicated in the statement.

6.4 Ordering Data - Procurement documents should specify the levels of preservation, packaging, and packing required. (See 5.1).

NOTICE: When Government drawings, specifications, or other data are furnished for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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